

$$\begin{aligned}
 42. \quad L(p) &= \prod_{i=1}^n p (1-p)^{y_i-1} \\
 &= \tilde{p}^n (1-p)^{-n + \sum_{i=1}^n y_i} \\
 &= \underbrace{\tilde{p}^n (1-p)^{-n + n\bar{y}}}_{g(\bar{y}, p)} \cdot \underbrace{1}_{h(y_1, \dots, y_n)}
 \end{aligned}$$

hence by Factorization Criterion, $\bar{y}$ is sufficient for p . ✓

$$\begin{aligned}
 43. \quad L(\alpha) &= \prod_{i=1}^n \alpha y_i^{\alpha-1} \div y_i \div \theta^{\alpha} \\
 &= \underbrace{\alpha^n \prod_{i=1}^n y_i^{\alpha-1} \cdot \left(\prod_{i=1}^n y_i \right)^{-1}}_{g(\bar{y}, \alpha)} \cdot \theta^{-n\alpha}
 \end{aligned}$$

hence by Factorization Criterion, $\sum_{i=1}^n y_i$ is sufficient for α .
 $\underbrace{\quad}_{h(y_1, \dots, y_n)}$

$$4.4 \quad L(\alpha) = \prod_{i=1}^n \alpha^{\alpha} y_i^{-(\alpha+1)}$$

$$= (\alpha^{\alpha})^n \div \underbrace{\left(\prod_{i=1}^n y_i^{\alpha} \cdot \prod_{i=1}^n y_i \right)}_{g(\underline{y}, \alpha)} \cdot \underbrace{1}_{h(\underline{y}, \alpha)}$$

hence by Factorization Criterion $\prod_{i=1}^n y_i$ is sufficient for α

$$5.1 \quad f(y|\theta) = \begin{cases} e^{-(y-\theta)} & y \geq \theta \\ 0 & \text{o.w.} \end{cases}$$

$$\begin{aligned} L(\theta) &= \prod_{i=1}^n e^{-(y_i-\theta)} \cdot 1_{\{y_i \geq \theta\}} \\ &= \prod_{i=1}^n e^{-y_i} \cdot e^{\theta} \cdot 1_{\{y_i \geq \theta\}} \\ &= e^{-\sum_{i=1}^n y_i} \cdot e^{n\theta} \cdot 1_{\{y_{(n)} \geq \theta\}} \\ &= \underbrace{e^{-\sum_{i=1}^n y_i}}_{g(\underline{y}, \theta)} \cdot \underbrace{e^{n\theta} \cdot 1_{\{y_{(n)} \geq \theta\}}}_{h(\underline{y}, \theta)} \end{aligned}$$

hence by Factorization Criterion, $Y_{(n)}$ is sufficient for θ

$$5.2 \quad L(\theta) = \prod_{i=1}^n \frac{1}{\theta^2} \cdot y_i^2 \cdot 1_{\{0 \leq y_i \leq \theta\}}$$

$$= \frac{1}{\theta^{2n}} \cdot \prod_{i=1}^n y_i^2 \cdot 1_{\{0 \leq y_i \leq \theta\}}$$

$$= \frac{1}{\theta^{2n}} \cdot \prod_{i=1}^n y_i^2 \cdot 1_{\{0 \leq y_{(n)} \leq \theta\}} \cdot 1_{\{y_{(n)} \leq \theta\}}$$

$$= \underbrace{\frac{1}{\theta^{2n}} \cdot 1_{\{y_{(n)} \leq \theta\}}}_{g(\underline{y}_{(n)}, \theta)} \cdot \underbrace{\left(\prod_{i=1}^n y_i^2 \right)}_{h(\underline{y}, \dots, y_{(n)})} \cdot 1_{\{0 \leq y_{(n)} \leq \theta\}}$$

hence by Factorization Criterion, $Y_{(n)}$ is sufficient for θ